



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 09

Working Capital Model

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Homework 1

We extend the working capital model of Class 9. Households are unchanged: they hold cash and can also deposit funds N_t^i in a financial intermediary (bank), and firms borrow from the bank to finance their wage bill in advance. We add a cash-in-advance requirement on firms: a firm must enter period t holding real money balances m_t^f sufficient to cover at least 25% of the wage bill it intends to finance through borrowing. Since m_t^f is chosen in $t - 1$ (before period- t variables are known), the relevant benchmark is period $t - 1$'s realized wage bill.

Households

Households maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i, 1 - h_t^i)$$

subject to the cash-in-advance constraint

$$P_t c_t^i \leq M_{t-1}^i - N_t^i$$

and the budget constraint

$$\frac{M_t^i}{P_t} + k_{t+1}^i = w_t h_t^i + r_t k_t^i + (1 - \delta) k_t^i + \frac{r_t^n N_t^i}{P_t}$$

where N_t^i denotes nominal deposits at the financial intermediary and r_t^n is the gross deposit rate. This part of the model is unchanged from the baseline working capital setup.

Firms

Firms operate a Cobb–Douglas technology

$$Y_t = \lambda_t K_t^\theta H_t^{1-\theta}$$

and must finance their wage bill $w_t H_t$ before production takes place. Unlike the baseline model, firms are now required to hold real cash balances m_t^f satisfying

$$m_t^f \geq 0.25 w_{t-1} H_{t-1}$$

The remainder of the wage bill is financed through a loan from the bank at the gross rate r_t^f . Denoting the real loan amount $\ell_t \equiv w_t H_t - m_t^f$, the firm's period- t budget constraint is

$$Y_t = r_t^f \ell_t + m_t^f + r_t K_t$$

i.e. the firm repays r_t^f on the borrowed portion, repays (uses) its own cash m_t^f one-for-one, and pays the rental rate r_t on capital. Since firms are competitive and take λ_t, K_t as given each period, profit maximization with respect to H_t and K_t yields

$$r_t = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta} = \theta Y_t / K_t$$

$$r_t^f w_t = (1 - \theta) \lambda_t K_t^\theta H_t^{-\theta} = (1 - \theta) Y_t / H_t$$

which are the same factor-market conditions as in the baseline model: the CIA requirement affects the firm's financing mix, not its static marginal-product conditions, since m_t^f is predetermined at the time the firm chooses H_t and K_t .

Financial intermediaries

Banks take deposits N_t from households and lend ℓ_t (in real terms) to firms, plus intermediate the money injection $(g_t - 1)M_{t-1}$ from the central bank. The zero-profit condition is

$$r_t^f (N_t + (g_t - 1)M_{t-1}) = r_t^n N_t$$

Credit market clearing now differs from the baseline: since firms cover part of the wage bill with their own cash, the loanable funds needed equal only the borrowed portion of the wage bill,

$$N_t + (g_t - 1)M_{t-1} = P_t (w_t H_t - m_t^f)$$

Exogenous processes and market clearing

Money supply follows

$$M_t = g_t M_{t-1}$$

and the aggregate resource constraint (goods market clearing) is

$$Y_t = C_t + K_{t+1} - (1 - \delta)K_t$$

Summary of new elements relative to the baseline model

- New firm state variable: m_t^f , real cash held by the firm entering period t .
- New constraint: $m_t^f \geq 0.25 w_{t-1} H_{t-1}$.
- Firm's loan demand shrinks from $w_t H_t$ to $\ell_t = w_t H_t - m_t^f$.
- Credit market clearing condition is adjusted accordingly.

Homework 2

We keep using the previous working capital model, but we now require banks to hold a fraction $\tau = 0.20$ of the total funds at their disposal as reserves. In each period the financial intermediary receives deposits N_t from households plus a money transfer of $(g_t - 1)M_{t-1}$ from the central bank, so total funds are $N_t + (g_t - 1)M_{t-1}$. Reserves are held as cash inside the period, pay no interest, and are returned at the end of the period; only the remaining fraction $1 - \tau$ can be lent to firms.

Households, firms, the money supply rule and the resource constraint are exactly as previously. Only the two equations describing the banking sector change.

Financial intermediaries

Banks lend $(1 - \tau)(N_t + (g_t - 1)M_{t-1})$ at the gross rate r_t^f and hold $\tau(N_t + (g_t - 1)M_{t-1})$ in reserves, which return one for one. The zero profit condition is therefore

$$\left[\tau + (1 - \tau) r_t^f \right] (N_t + (g_t - 1)M_{t-1}) = r_t^n N_t,$$

and the credit market clears when the amount actually lent equals the nominal wage bill,

$$(1 - \tau)(N_t + (g_t - 1)M_{t-1}) = P_t w_t H_t.$$

Stationary state

With $\bar{r} = 1/\beta - 1 + \delta$ and $\bar{r}^n = \bar{g}/\beta$, the stationary state solves five equations in $\bar{M}/\bar{P}$, $\bar{N}/\bar{P}$, $\bar{r}^f$, $\bar{C}$ and $\bar{H}$:

$$\bar{r}^f = - \frac{\beta(1 - \theta) \left(\frac{\theta}{\bar{r}}\right)^{\frac{\theta}{1-\theta}}}{\bar{C}\bar{g}B}, \quad (1)$$

$$\left[\tau + (1 - \tau)\bar{r}^f \right] \left(\bar{N}/\bar{P} + \left[1 - \frac{1}{\bar{g}} \right] \bar{M}/\bar{P} \right) = \bar{r}^n \bar{N}/\bar{P}, \quad (2)$$

$$- \frac{\bar{C}\bar{g}B}{\beta} \bar{H} = (1 - \tau) \left(\bar{N}/\bar{P} + \left[1 - \frac{1}{\bar{g}} \right] \bar{M}/\bar{P} \right), \quad (3)$$

$$\bar{C} = \frac{\bar{M}/\bar{P}}{\bar{g}} - \bar{N}/\bar{P}, \quad (4)$$

$$\bar{M}/\bar{P} = \frac{\bar{g}}{\beta} \bar{N}/\bar{P} + \left[(\bar{r} - \delta) \left(\frac{\theta}{\bar{r}}\right)^{\frac{1}{1-\theta}} - \frac{\bar{C}\bar{g}B}{\beta} \right] \bar{H}. \quad (5)$$

Equations (2) and (3) are the ones modified by the reserve requirement; (1), (4) and (5) are the same as in Class 9. Note that $\bar{r}$ and $\bar{r}^n$ do not involve τ , so the real return on capital and the deposit rate are unchanged.

Results

The system is solved numerically for $\beta = 0.99$, $\delta = 0.025$, $\theta = 0.36$, $B = -2.5806$ and $\bar{\lambda} = 1$.

Case 1: no reserve requirement ($\tau = 0$)

Annual inflation	−4%	0	10%	100%	400%
$\bar{g}$	0.9900	1.0000	1.0240	1.1900	1.4100
$\bar{r}$	0.0351	0.0351	0.0351	0.0351	0.0351
$\bar{r}^n$	1.0000	1.0101	1.0343	1.2020	1.4242
$\bar{r}^f$	1.0221	1.0101	0.9824	0.8259	0.6820
$\overline{M/P}$	1.6556	1.6675	1.6960	1.8895	2.1395
$\overline{N/P}$	0.7735	0.7672	0.7522	0.6625	0.5716
$\bar{C}$	0.8988	0.9003	0.9040	0.9253	0.9458
$\bar{Y}$	1.2087	1.2108	1.2157	1.2444	1.2719
$\bar{w}$	2.3193	2.3469	2.4130	2.8702	3.4762
$\bar{H}$	0.3263	0.3269	0.3282	0.3359	0.3434
$\bar{K}$	12.3963	12.4180	12.4686	12.7622	13.0449
utility	−0.9488	−0.9485	−0.9479	−0.9446	−0.9419

Case 2: 20% reserve requirement ($\tau = 0.20$)

Annual inflation	−4%	0	10%	100%	400%
$\bar{g}$	0.9900	1.0000	1.0240	1.1900	1.4100
$\bar{r}$	0.0351	0.0351	0.0351	0.0351	0.0351
$\bar{r}^n$	1.0000	1.0101	1.0343	1.2020	1.4242
$\bar{r}^f$	1.0247	1.0126	0.9849	0.8280	0.6837
$\overline{M/P}$	1.8378	1.8523	1.8868	2.1223	2.4281
$\overline{N/P}$	0.9599	0.9542	0.9408	0.8604	0.7786
$\bar{C}$	0.8965	0.8981	0.9018	0.9230	0.9434
$\bar{Y}$	1.2057	1.2078	1.2127	1.2413	1.2687
$\bar{w}$	2.3136	2.3410	2.4070	2.8631	3.4675
$\bar{H}$	0.3255	0.3261	0.3274	0.3351	0.3425
$\bar{K}$	12.3653	12.3871	12.4375	12.7304	13.0124
utility	−0.9492	−0.9489	−0.9483	−0.9449	−0.9422

Comparison of the two stationary states

- Every real quantity — $\bar{C}$, $\bar{H}$, $\bar{K}$, $\bar{Y}$, $\bar{w}$ — falls by 0.25%, and by the same 0.25% at every inflation rate. The system implies $\bar{w}(\tau)/\bar{w}(0) = (1 - \tau)/(1 - \beta\tau) = 0.8/0.802$, and all of them scale by that factor, so $\bar{K}/\bar{H}$, $\bar{Y}/\bar{H}$ and $\bar{C}/\bar{Y}$ are unchanged.
- The effect is this small because reserves are held for one period only. What they cost is about one period of interest ($1 - \beta \approx 1\%$) on a fifth of the bank's funds.
- $\bar{r}^f$ rises by that same 0.25%, since $\bar{r}^f \bar{w}$ is pinned down by the capital-labour ratio. $\bar{r}$ and $\bar{r}^n$ do not move at all: the requirement simply widens the spread between what banks charge and what they pay.
- The large change is nominal. $\overline{N/P}$ rises 24% and $\overline{M/P}$ rises 11% at zero inflation. For the same wage bill to get financed, households have to deposit roughly $1/(1 - \tau)$ times as much. Since $\bar{M}$ is exogenous, this is a one-time fall in the price level.
- Welfare is lower with the requirement in every column, though barely: -0.9489 against -0.9485 at zero inflation.
- Inflation still does the same thing. Higher $\bar{g}$ raises $\bar{H}$, $\bar{Y}$ and $\bar{C}$ and lowers $\bar{r}^f$, exactly as before. The requirement shifts the table down a little; it does not change how the economy responds to money growth.